

Multimedia Application

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Content

- ▶ LLM
- ▶ Transformer
- ▶ Pre-training
- ▶ Finetuning
- ▶ SFT
- ▶ Preference tuning

LLM

- ▶ LLM = Large Language Model
- ▶ "A language model is a statistical or machine learning model that **assigns probabilities** to **sequences of tokens**."

Large

- ▶ Model size: billions of parameters or more
- ▶ Training data: 100s of billions of tokens or more
- ▶ Compute: a lot of GPUs

LLM pipeline

- ▶ We don't just "program" an AI to be smart; we grow it through stages. Think of it like training a chef:
- ▶ **Pre-training:** Reading every cookbook and food blog in existence (General knowledge).
- ▶ **Fine-tuning:** Learning how to actually take an order and cook a specific meal (Instruction following).
- ▶ **Preference Tuning:** Learning that some people hate cilantro or prefer their steak medium-rare (Human values/safety).

Pre-training

- ▶ This is the most computationally expensive phase. The goal is to build a "Base Model."
- ▶ The Data: Trillions of tokens from the internet (Wikipedia, Books, GitHub, ArXiv).
- ▶ The Objective: Self-Supervised Learning. The model plays a constant game of "Predict the Next Token."
- ▶ The Result: The model learns grammar, world facts, reasoning, and even some coding.
- ▶ The Catch: A pre-trained model isn't an assistant. If you ask it "What is the capital of France?", it might respond with "And what is the capital of Germany?" because it thinks it's completing a quiz list, not answering a question.

Fine-tuning

- ▶ This stage turns a "Base Model" into an "Instruct Model." We use Supervised Fine-Tuning (SFT).
- ▶ The Data: A smaller, high-quality dataset of (Prompt, Response) pairs written by humans.

Prompt: "Write a poem about a toaster.

Response: "O metal box of silver hue.

- ▶ **The Goal:** Teach the model the format of a conversation. It learns that when a human asks a question, the correct "next tokens" should be a direct answer.
- ▶ **Limitation:** SFT is great for format, but it's hard to scale human writing for every possible nuanced scenario or safety edge case.

Preference Tuning

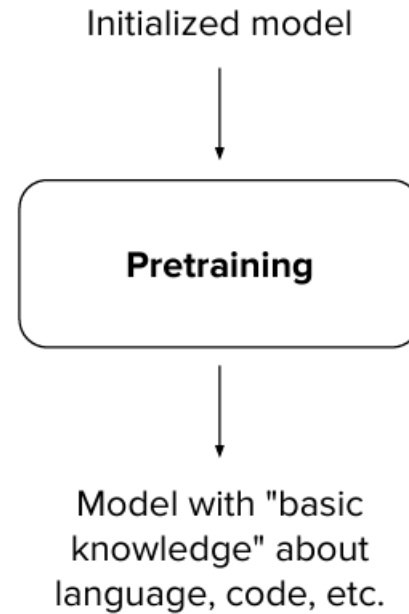
- ▶ Even after fine-tuning, a model might be rude, biased, or confidently wrong. Preference tuning aligns the model with human values.
- ▶ RLHF (Reinforcement Learning from Human Feedback):
- ▶ The model generates two different answers to the same prompt.
- ▶ A human ranks them (Answer A is better than Answer B).

Preference Tuning

- ▶ A Reward Model is trained to predict what humans like.
- ▶ The LLM is updated to maximize its "reward" score.
- ▶ DPO (Direct Preference Optimization): A newer, more efficient method that skips the separate Reward Model and optimizes the LLM directly on the ranked data.
- ▶ The Goal: Ensuring the model is "Helpful, Honest, and Harmless"

What is Pretraining?

- ▶ Initial training phase
- ▶ Uses large-scale unlabeled data
- ▶ Learns general language patterns

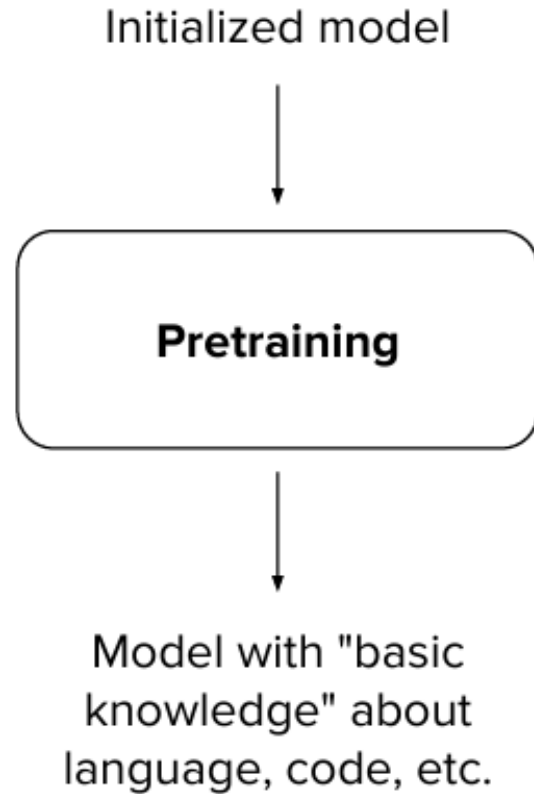


Pretraining Objectives

- ▶ Predict next word (autoregressive)
- ▶ Masked language modeling
- ▶ Learn grammar, facts, reasoning

Pretraining Data

- ▶ Web text, books, articles
- ▶ Billions of tokens
- ▶ Diverse domains



Advantages of Pretraining

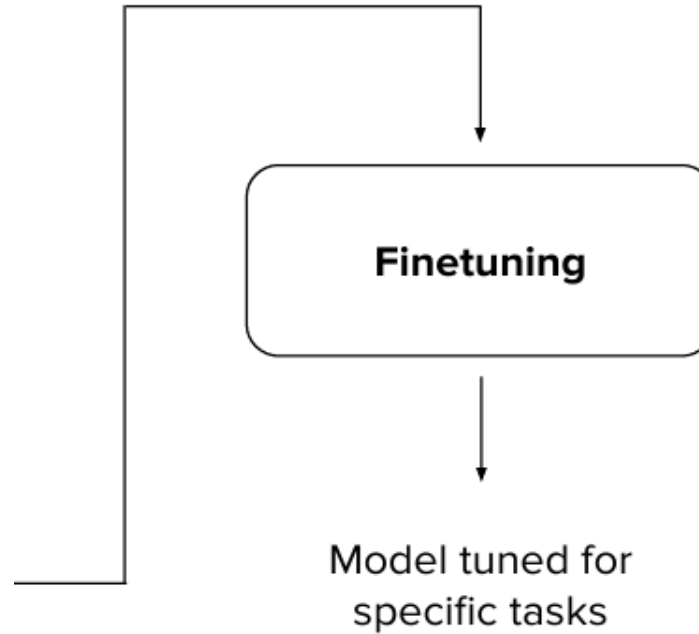
- ▶ General knowledge
- ▶ Transfer learning capability
- ▶ Strong base model

Limitations of Pretraining

- ▶ Not task-specific
- ▶ Can generate unsafe outputs
- ▶ Needs alignment

What is Fine-Tuning?

- ▶ Training on labeled data
- ▶ Focus on specific tasks
- ▶ Improves performance

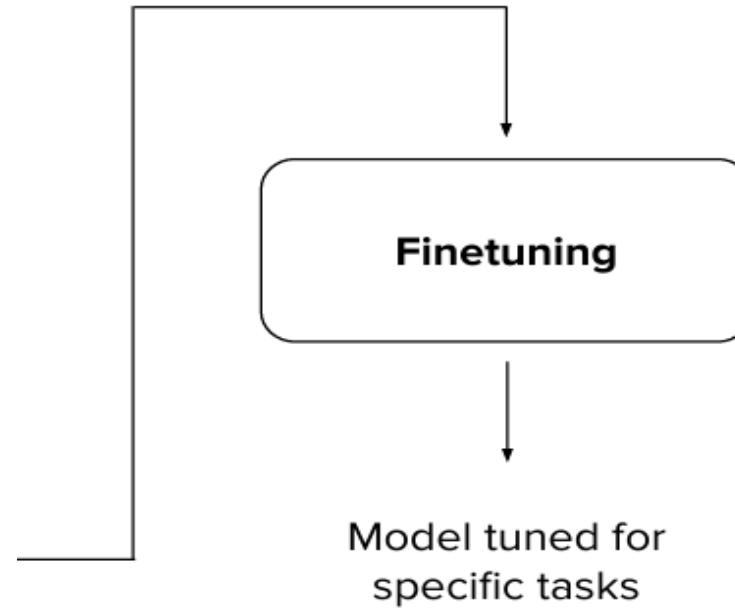


Fine-Tuning Examples

- ▶ Chatbots
- ▶ Medical QA
- ▶ Code generation

Fine-Tuning Process

- ▶ Start with pretrained model
- ▶ Use smaller dataset
- ▶ Adjust weights



Types of Fine-Tuning

- ▶ Supervised fine-tuning (SFT)
- ▶ Instruction tuning
- ▶ Domain adaptation

Supervised fine-tuning (SFT)

- ▶ Change the model behavior by tuning its weights
- ▶ Strategy.
- ▶ Collect pairs of input/outputs with desired behavior (also known as SFT data)
- ▶ Train using next word prediction objective given the input

Special case. SFT on instruction-following data: "Instruction tuning"

Instruction tuning overview

- Goal. "Graduate" the model to being a helpful assistant

Story writing

Write a short story about a teddy bear who likes to read poetry.

Once upon a time, a bear, Teddy, stumbled upon verses from Attar. The bear became so captivated with the rhythm of the poem's sentences that he got filled with happiness.

Poem creation

Create a poem about my cute teddy bear.

Soft and cuddly, full of charm,
Always keeps me safe from harm,
With button eyes and fur so neat,
My teddy bear is oh so sweet.

Lists generation

List three fun activities a teddy bear might do on a rainy day.

Sure!

1. Read poetry with friends.
2. Be cute.
3. Hug its owner tightly.

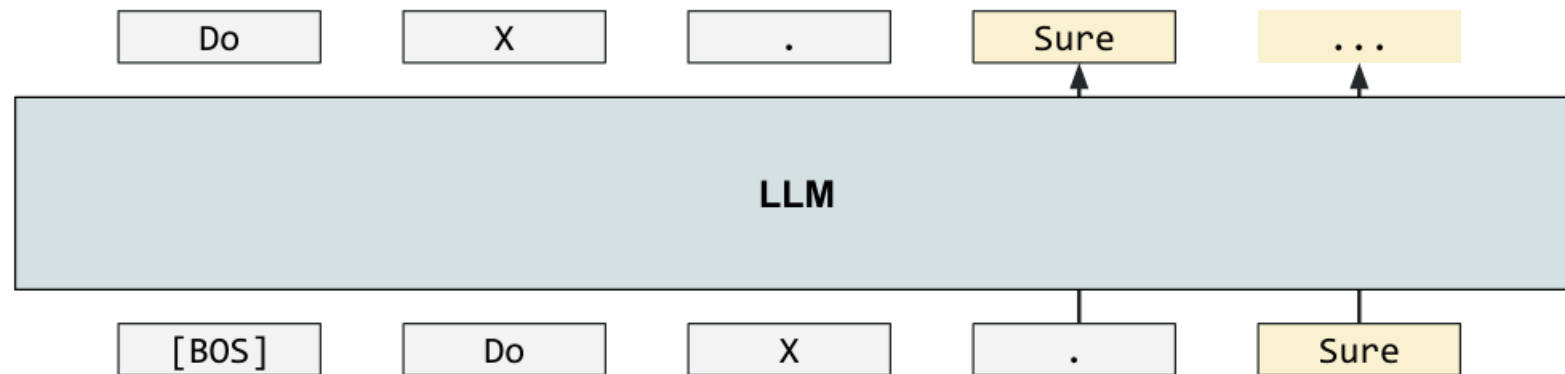
Explanation

Explain why a teddy bear is a great friend.

A teddy bear is a great friend because it provides comfort and companionship. Its soft fur and cuddly nature is perfect for hugs. Teddy bears are always there in both happy and sad moments.

Instruction tuning overview

- ▶ Objective function. Predict next token given the input.



Instruction tuning overview

- ▶ Data mixtures.
- ▶ Can be both human-written and synthetic data.
- ▶ Assistant dialogs
- ▶ Synthetic instructions
- ▶ Maths, reasoning, code
- ▶ Safety alignment

Instruction tuning overview

Size. Thousands to millions of examples.

Model	SFT size (# examples)
GPT-3	13 thousands
LLaMA 3	10 million

Behavior example

Can I put my
teddy bear in
the washer?



**Pretrained +
instruction
tuned LLM**



No, it might get
damaged. Try hand
washing instead.

Challenges

- ▶ Very high-quality data needed
- ▶ Sensitive to prompt distribution
- ▶ Generalization
- ▶ Difficult to evaluate
- ▶ Computationally expensive

Benchmarks

- ▶ Dimensions.
- ▶ General knowledge: MMLU
- ▶ Basic reasoning: ARC-Challenge
- ▶ Math reasoning: GSM8K
- ▶ Code generation: HumanEval

Benefits of Fine-Tuning

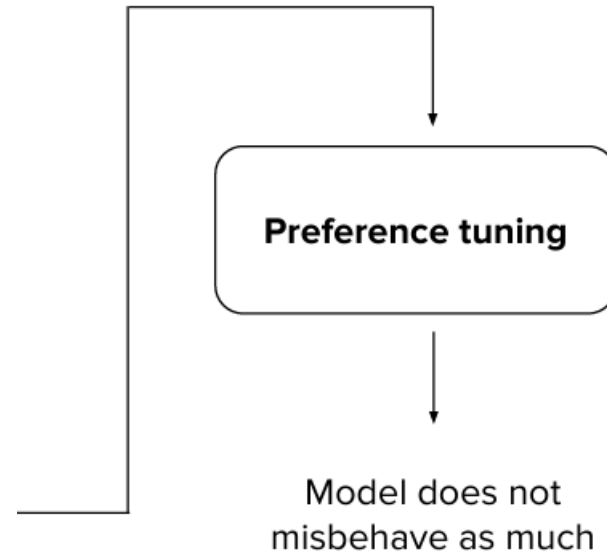
- ▶ Task specialization
- ▶ Better accuracy
- ▶ Controlled outputs

Challenges in Fine-Tuning

- ▶ Overfitting
- ▶ Data quality issues
- ▶ Cost

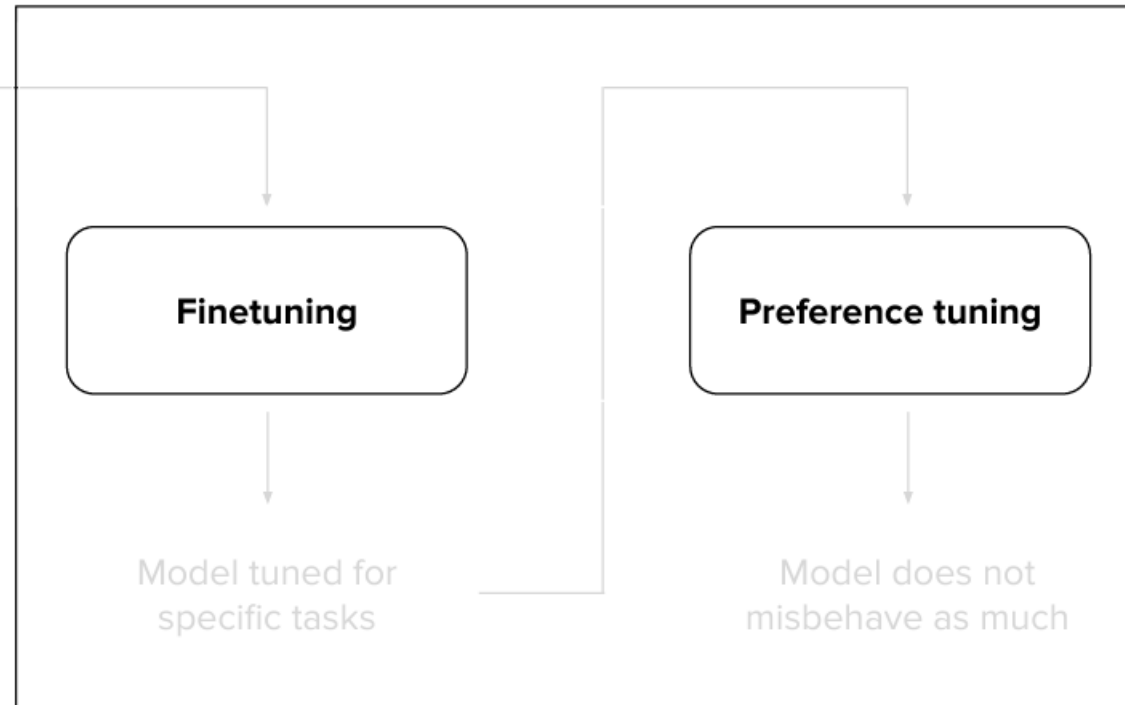
What is Preference Training?

- ▶ Align model with human values
- ▶ Uses human feedback
- ▶ Improves responses



Human Feedback

- ▶ Ranking responses
- ▶ Choosing better output
- ▶ Reinforcement signals



"Alignment" of the model

RLHF (Reinforcement Learning from Human Feedback)

- ▶ Reward model
- ▶ Policy optimization
- ▶ Iterative improvement

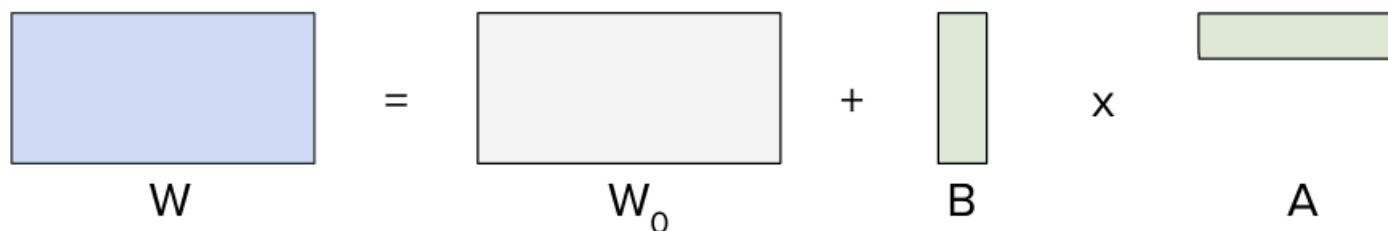
Preference Training Methods

- ▶ RLHF
- ▶ Direct Preference Optimization (DPO)
- ▶ Constitutional AI

Parameter-efficient finetuning with LoRA

- Context. SFT is resource intensive and not everyone has big GPUs.

Idea. Low-Rank Adaptation (LoRA) approximates matrix with product of two low-rank matrices


$$W = W_0 + B \times A$$

Fraction of parameters need to be trained with similar performance
Other methods include prefix tuning and adapters

Why Preference Training?

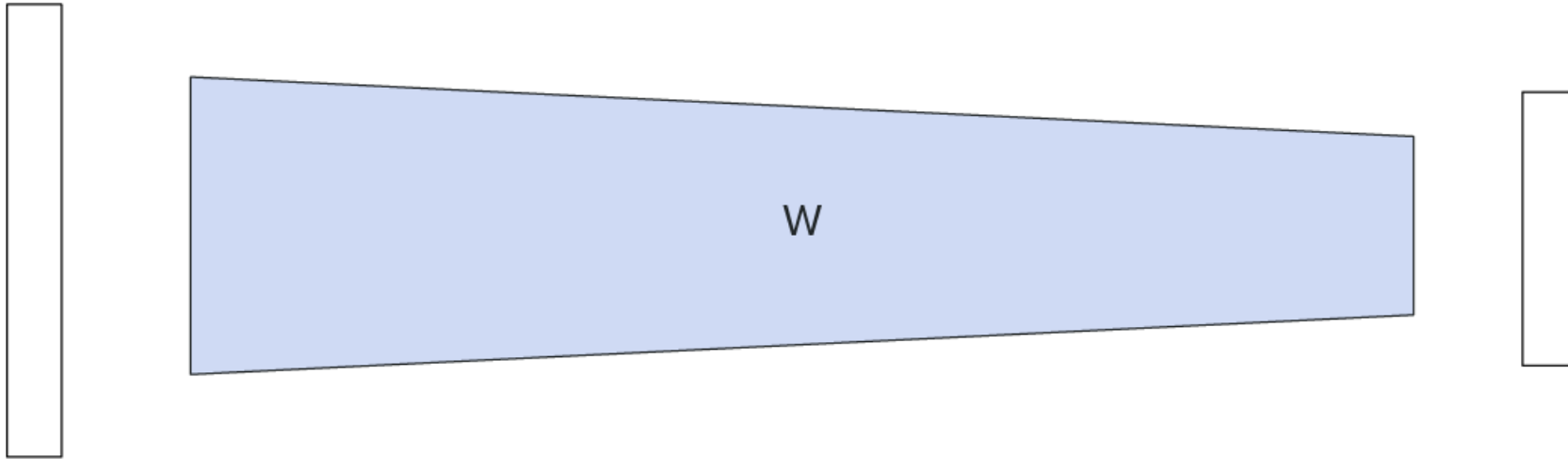
- ▶ Safety
- ▶ Helpfulness
- ▶ Alignment with user expectations

Pipeline Overview

- ▶ Pretraining → Fine-tuning → Preference training
- ▶ Progressive improvement

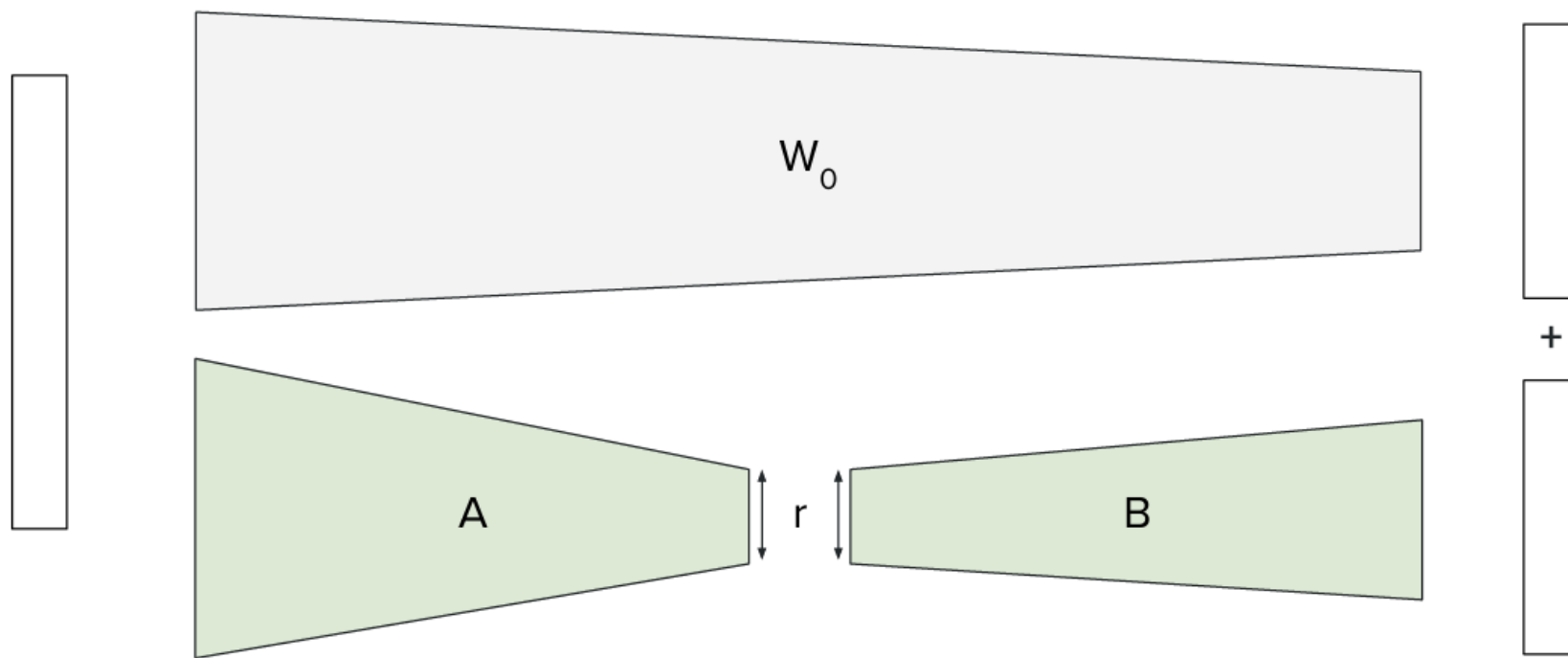
Finetuning evolution

Before. "Regular" finetuning optimizes the full matrix of weights

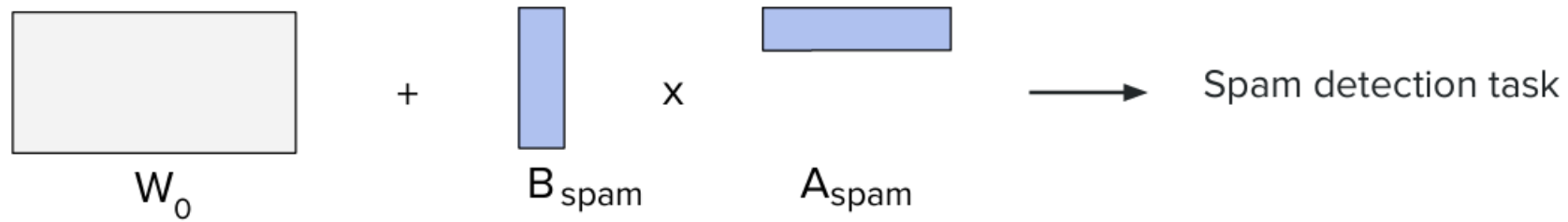


Finetuning evolution

After. LoRA finetuning optimizes the full matrix of weights



Benefit of LoRA: swap matrices = swap tasks



Training dynamics

- ▶ Beware of training differences:
- ▶ LoRA needs a higher learning rate than full fine-tuning
- ▶ LoRA does poorly on large batch size compared to full fine-tuning

Real-World Example

- ▶ GPT models
- ▶ Chat assistants
- ▶ Search engines

Challenges in Full Pipeline

- ▶ Cost and compute
- ▶ Bias and fairness
- ▶ Evaluation difficulty

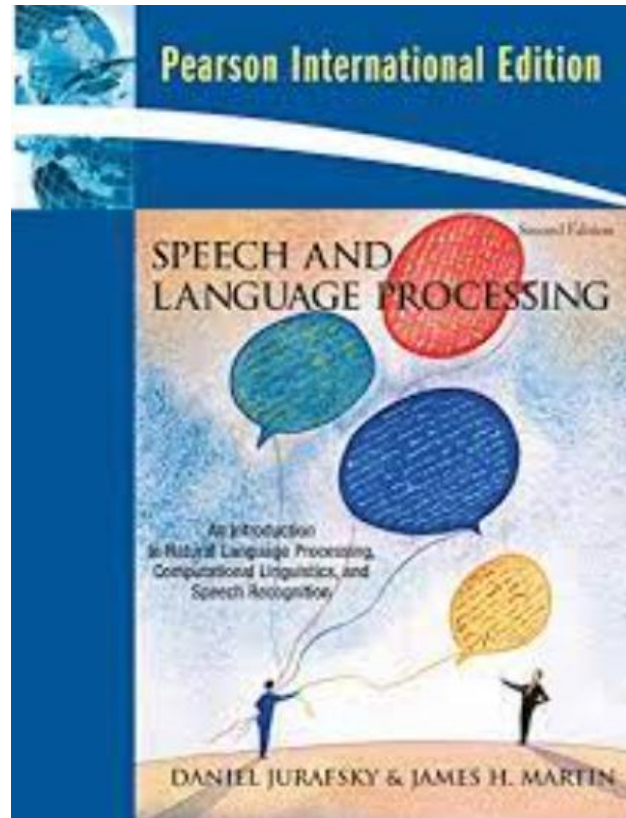
Future Trends

- ▶ Smaller efficient models
- ▶ Better alignment methods
- ▶ Multimodal learning

Summary

- ▶ Pretraining builds knowledge
- ▶ Fine-tuning adds specialization
- ▶ Preference training aligns behavior

Reference



Chapter 10

Question



Thank you

